

“Phase I” method for initial feasible point (0.25 pt)

We consider the original constraints:

$$\begin{aligned}(\text{C1}) \quad & x - 2y + 2 \geq 0, \\(\text{C2}) \quad & -x - 2y + 6 \geq 0, \\(\text{C3}) \quad & -x + 2y + 2 \geq 0, \\(\text{C4}) \quad & x \geq 0, \\(\text{C5}) \quad & y \geq 0.\end{aligned}$$

Given the starting point

$$x_0 = \begin{pmatrix} 1 \\ 2.5 \end{pmatrix},$$

we first check feasibility:

$$\begin{aligned}\text{C1: } & 1 - 2 \cdot 2.5 + 2 = -2 \quad (\text{violated by } 2), \\ \text{C2: } & -1 - 2 \cdot 2.5 + 6 = 0 \quad (\text{active}), \\ \text{C3: } & -1 + 2 \cdot 2.5 + 2 = 6 \quad (> 0), \\ \text{C4: } & 1 \geq 0, \quad \text{C5: } 2.5 \geq 0.\end{aligned}$$

Only Constraint (C1) is violated.

Phase I reformulation

Introduce the slack variable $s \geq 0$ to account for infeasibility and modify the constraints to follow the convention used in the lecture:

$$\begin{aligned}\min_{x,y,s} \quad & s \\ \text{s.t.} \quad & -x + 2y - 2 \leq s, \\ & x + 2y - 6 \leq 0, \\ & x - 2y - 2 \leq 0, \\ & -x \leq 0, \quad -y \leq 0, \quad s \geq 0.\end{aligned}$$

Choose $s_0 = 2 = \max(0, g_1(x_0))$ so that Constraint (C1) becomes satisfied at the starting point:

$$-1 + 2 \cdot 2.5 - 2 = 2.$$

Thus $(x_0, y_0, s_0) = (1, 2.5, 2)$ is feasible for the Phase I problem.

Solving Phase I

Because the original problem is feasible, the minimum of Phase I is $s^* = 0$, attained at any point satisfying (C1)–(C5).

A suitable feasible point is found by intersecting (C1) and (C2):

$$\begin{cases} -x + 2y - 2 = 0, \\ x + 2y - 6 = 0. \end{cases}$$

Solve the system:

$$\begin{aligned}x &= 2y - 2, \\ (2y - 2) + 2y - 6 &= 0 \Rightarrow 4y - 8 = 0 \Rightarrow y = 2,\end{aligned}$$

$$x = 2 \cdot 2 - 2 = 2.$$

Thus the candidate point is:

$$(x, y, s) = (2, 2, 0).$$

Check feasibility:

$$\text{C1: } 2 - 2 \cdot 2 + 2 = 0 \geq 0,$$

$$\text{C2: } -2 - 2 \cdot 2 + 6 = 0 \geq 0,$$

$$\text{C3: } -2 + 2 \cdot 2 + 2 = 4 \geq 0,$$

$$\text{C4: } 2 \geq 0, \quad \text{C5: } 2 \geq 0,$$

$$s = 0.$$

Thus, a valid feasible point for the original problem, produced by the Phase I method, is

$$x_{\text{feasible}} = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$$

which satisfies all original constraints and can be used to start the active-set method.